Topic 2: ICT risks and treat management

Module 2

IT project risk management

Introduction

This module focuses on identifying and managing the potential risk that could lead to IT development project failure. It examines the concept of doing a risk assessment to identify possible risks. This module also investigates common risks associated with IT development projects, and then explores the uses of a contingency plan as one way of dealing with project risk factors

Project risk: an event or failure that can impact the success of the project

Unit 2.1: Risk assessment and management for IT development projects

Definition: Risk assessment is the process require an analysis of the probability (likelihood) of a risk occurring, and determines how severe its impact would be on the project

2.1.1 Risk management

Definition: Risk management is involves planning, identifying potential risks, assessing the risks, putting in place a contingency plan to deal with any risks that occur, and tracking and controlling risks

Definition: Contingency Planning involves identifying ways of dealing with any risks that may arise. It usually involves identifying actions to take and the people responsibility for taking action.

Managing project risks

2.1.2 Risk assessment

Risk assessment is an analysis of the probability of a risk occurring (low - high). The project manager must use judgement, experience and expert advice to accurately

The manager uses the assessment of the identified risks to create a contingency plan. The higher the perceived risk (in terms of probability and impact)

2.1.3 implication of risk and threats to IT project development

Recall from level 3 course on system analysis and design (SAD L3 that project success is measured by ensuring that the project.

- Is delivered on time (meets the schedule
- Is under (or on) budget
- Delivers all functionality to the users' satisfaction

Project risks are the things that are likely to cause problems with one or more of the above. It is therefore very important that a project manager takes risk management seriously, and has a knowledge and skills to identify, assess and mitigate for risks using a contingency plan

Mitigate to prepare for and lessen the impact of risk

Unit 2.2: Preventing IT project failure

This unit examines common reasons for IT project failure. It then explores how to identify the risk factors linked to these in order to assess the risks and manage them successfully.

2.2.1 Common reasons for the failure of projects

The CHAOS report looks at the percentage of project that can be classified as successful according to whether they meet three main requirements:

- Delivered on time
- Delivered within budget
- Meeting scope

CHAOS report show:

- Successful project (project meeting all three requirement), challenged project (project completed but not meeting all three requirements), and failed/impaired project (projects abandoned or cancelled).
- Comparison between different sized projects, various complexities of projects, and different methodologies used, for example agile vs traditional structured waterfall (refer back to SAD L3 to refresh your memory about these approaches).

It is common for the project manager to use the above factors to guide them in identifying project risks that fall into these areas. Experience from previous project and PMBOK (the project management body of knowledge) are also helpful

2.2.2 Identifying major risk factors for projects

Identifying the major risk factors for a project is done by looking at the type of challenges identified in the CHAOS or similar reports

Factors

- Incomplete requirements
- Lack of trainers
- Additional hardware upgrades

Unit 2.3: Contingency planning for IT project

Contingency plan takes place after risk factors have been identified, assessed and added to the risk register or similar tracking document. Contingency planning involves planning for future scenarios. In the case of IT project development it involves identifying ways to deals with any risk that may arise. This is known as risk mitigation

2.3.1 Risk mitigation: actions that can be taken relevant to various identified risks and threat

Risk mitigation can take one of four approaches depending on the circumstances and context. Different actions are required to deal with different types of risks

The four approaches of risk mitigation

Figure 2.7

Avoidance	• Attempts to prevent risk with other parties • Best strategy for high – risk factors, i.e. high probability and high damage, critical path tasks
Transfer	• Shares the risk with other parties • Used for risks with low probability of occurring • Example: if the risk involves possible increases in licence fees it might be worth negotiating a slightly higher payment with the supplier now rather than risk a big increase later
Acceptance	• Accepts risks • Used for risks with low potential damage. • Sometimes the cost of mitigation a small risk is higher than it is worth
Limitation	• Takes some action to prevent or lessen damage. • Typically involves a mix of avoidance and acceptance

Keywords and definition of the Topic 2

Keywords

1. Project risk
2. Mitigate

Definitions

1. Risk assessment
2. Risk management
3. Contingency planning

Test yourself on module 2

1. What is the difference between risk management and risk assessment? (4)
2. Explain what is important to manage risks in IT development projects. (3)
3. List and briefly explain three reasons for the failure of projects. (6)
4. Describe the purpose of contingency planning. (2)
5. Identify and discuss the four approaches to risk mitigation, and describe when each should be used. (12)

TOTAL: [27]